

SYSTEM ALIGNMENT REPORT

Da Bali Resort — Build vs. Blueprint

A comparison of the current Da Bali Resort web application against the capstone specification, *“Da Bali Resort Digital Transformation: Web-Based Booking and Reservation Management System”*

1. Overview

The current Da Bali Resort system was reviewed against the features and requirements listed in the capstone specification.

Assessment Scope

Item	Details
Method	Direct source code inspection
Files Reviewed	<code>server.ts</code> , <code>App.tsx</code> , <code>types.ts</code> , <code>schema</code>
Current Technology Stack	React · Express · SQLite
Documented Technology Stack	Vue · Node.js · MySQL
Current User Roles	Guest · Staff · Admin
Roles Specified in Capstone	Guest · Front-Desk · Housekeeping · Admin
Features Evaluated	24
Finished	15
Needs Improvement	4
Not Yet Built	5

The core booking and payment functions are already working well. However, several features from the specification are still missing, particularly the **FAQ chatbot**, **Housekeeping module**, **amenity inventory tracking**, **occupancy analytics**, and **audit logs**.

A Staff dashboard issue was also found and should be fixed before further development.

2. Authentication & User Access

The system supports secure passwords, user roles, sessions, and password recovery.

Feature	Status	Summary
Password Security	✅ Finished	Uses Bcrypt for password hashing.
Role-Based Access	⚠ Needs Improvement	Guest and Admin work, but the Staff dashboard has a routing issue.
Sessions	⚠ Needs Improvement	Uses server-side sessions instead of the JWT specified in the paper.
Password Recovery	✅ Finished	Includes email-based password reset.

Staff Dashboard Issue

Staff users currently encounter a blank dashboard after logging in.

The main cause is an outdated redirect:

Current: `dtr`

Correct: `staff-records`

The Staff role is also restricted from several API functions that currently require Admin access.

3. Booking & Availability

This is one of the strongest parts of the system.

Feature	Status	Summary
Date Conflict Checking	✅ Finished	Prevents overlapping bookings on the server.
Accommodation Booking	✅ Finished	Supports rooms, villas, cottages, guests, and extra beds.
Amenity Booking	✅ Finished	Amenities have their own booking and availability checks.

Real-Time Updates	✅ Finished	WebSockets update open dashboards automatically.
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Assessment

The booking system successfully addresses the main goal of preventing double-bookings.

4. Payments & Invoices

Feature	Status	Summary
Receipt Upload	✅ Finished	Guests can upload payment screenshots and references.
Payment Verification	✅ Finished	Staff/Admin can approve or reject payments.
Balance Tracking	✅ Finished	Tracks partial payments and remaining balances.
PDF Invoice	⚠️ Needs Improvement	Uses browser printing instead of generating an actual PDF.

5. Front-Desk & POS

Feature	Status	Summary
Walk-In Booking	✅ Finished	Supports immediate reservations and payment.
Booking Status	✅ Finished	Supports pending, confirmed, checked-in, completed, and no-show states.
Amenity Inventory	❌ Not Yet Built	Amenities currently have no quantity or stock tracking.

6. Housekeeping

The capstone specifies a dedicated Housekeeping role, but this has not yet been implemented.

Feature	Status
Housekeeping User Role	✗ Not Yet Built
Room Status	✗ Not Yet Built
Cleaning/Turnover Schedule	✗ Not Yet Built

7. Guest Communication

Feature	Status	Summary
FAQ Chatbot	✗ Not Yet Built	No chatbot currently exists.
Guest-to-Staff Messaging	✓ Finished	Guests can communicate with staff.
Automated Reminders	✓ Finished	Pre-arrival and post-checkout messages are supported.

8. Analytics & Reports

Feature	Status	Summary
Revenue Tracking	✓ Finished	Revenue totals are available on the dashboard.
Occupancy Analytics	✗ Not Yet Built	No room occupancy-rate calculation.
Report Export	⚠ Needs Improvement	CSV export currently focuses on staff attendance.
Audit Logs	✗ Not Yet Built	No persistent record of Admin/Staff actions.

9. Features Beyond the Original Specification

The current system also includes several features that were not explicitly required in the capstone specification.

- Staff Attendance
- Guest Feedback
- Homepage Content Management
- Automated Guest Communication
- Real-Time Synchronization

10. Additional Improvements

Aside from the features required by the capstone specification, the following improvements are recommended to make the system more secure, reliable, and user-friendly.

Improvement	Recommendation
UI/UX	Improve the overall design, consistency, responsiveness, and ease of use.
Guest Email Notifications	Send emails for booking confirmations, payment updates, cancellations, and reminders.
Authentication Rate Limiting	Add limits to repeated login and password-reset attempts.
Admin Calendar	Add a calendar view for reservations, check-ins, and check-outs.
Booking Search & Filters	Add easier searching and filtering by guest, date, room, status, and payment.
Mobile Responsiveness	Improve the experience on phones and tablets, especially for guests.

Admin Dashboard	Improve the dashboard with useful metrics such as bookings, revenue, occupancy, and pending tasks.
Activity History	Record important Admin and Staff actions for tracking and accountability.